AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 3 · STUDENT LESSON

Tokens and Context

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Before you start

AI-2 Week 25: tokenizer units need not equal words.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Account for fixed text, evidence and reserved output under a toy budget. Try the recall before opening notes.

Your learning target

I can account for fixed text, **evidence** and **reserved output** under a toy budget.

I can select complete **evidence** for a stated question.

I can handle insufficient or conflicting **evidence** explicitly.

Remember before reading

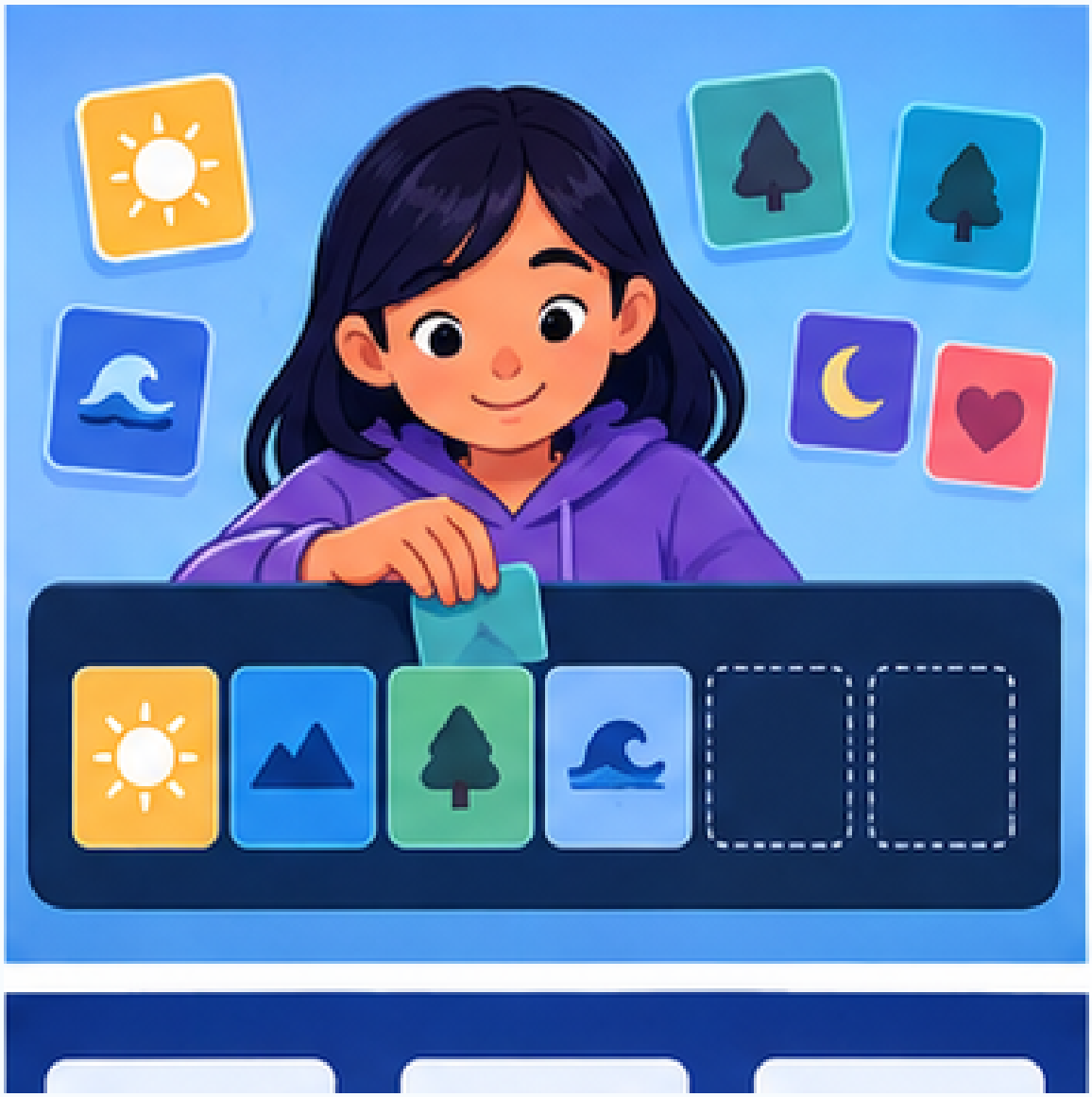
Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from A Distribution of Possibilities. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 3: **Token**s and Context

Models operate on **token**s within a limited **context window**.



CREATOR MISSION

Tokenize text, remove context, and predict failure changes.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Does lower next-**token** loss guarantee supported factual answers?

Readiness for Attention as weighted information selection

Why can equal row sums conceal different sequences?

My readiness explanation

This week’s end goal

By the end, I can budget context and preserve **evidence** needed to interpret a requested fact.

STUDENT LESSON · 2 OF 18

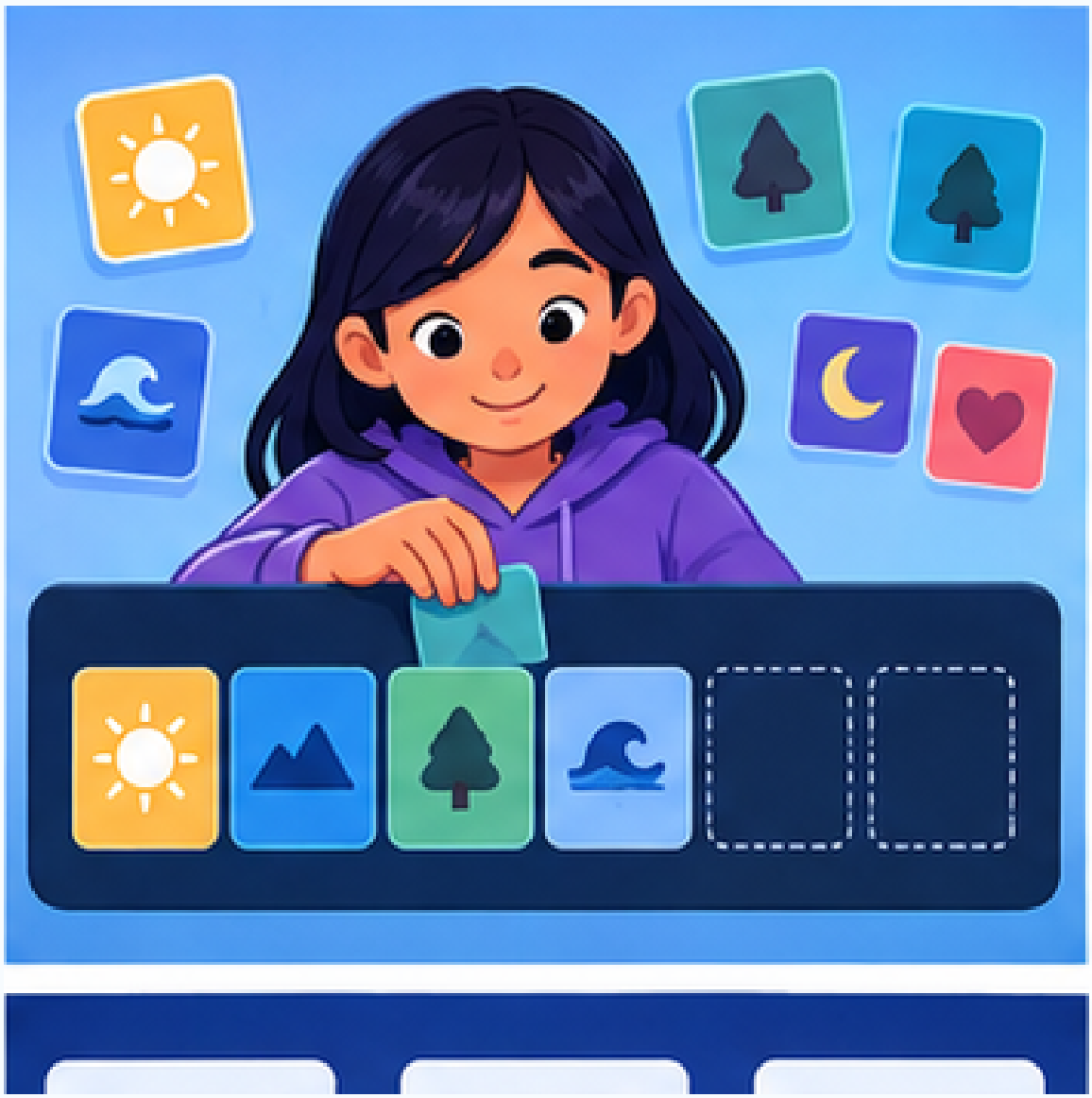
Notice the evidence

Investigate the supplied Tokens and Context case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

> The learners must use **token**s and context to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

For this lesson only, a fictional system allows 100 total **token** units for **input** plus **reserved output**. Its fixed instructions use 20 units, question 10 and **output** reserve 20. That leaves 50 units for **evidence**.

These are supplied counts, not word counts or a claim about any deployed model. Other systems can use different **input**/**output** accounting, so read their exact contract.

The question is: “Which room is approved for the robotics meeting, and what is its capacity?” **Evidence** A, a current approval notice, costs 30 units and names room B.

Evidence B, the current room register, costs 20 and gives room B capacity 12. **Evidence** C, an old approval notice, costs 25 and names room A.

Apply the connected idea: Attention as weighted information selection

Attention computes a weighted combination of **value** vectors, with weights derived from **query**-key compatibility. Queries, keys and **value**s are distinct roles even when they come from the same sequence.

A weight is a numerical contribution in this computation, not proof of the source's truth or permission. This week supplies weights directly; Week 7 calculates them from scores.

Use three **value**s V1=[2,0], V2=[0,4], V3=[4,2] with weights 0.5,0.25,0.25. **Value**s have equal width and weights are nonnegative with sum 1. Compute each coordinate separately. No **token** is literally copied by this combination.

STUDENT LESSON · 3 OF 18

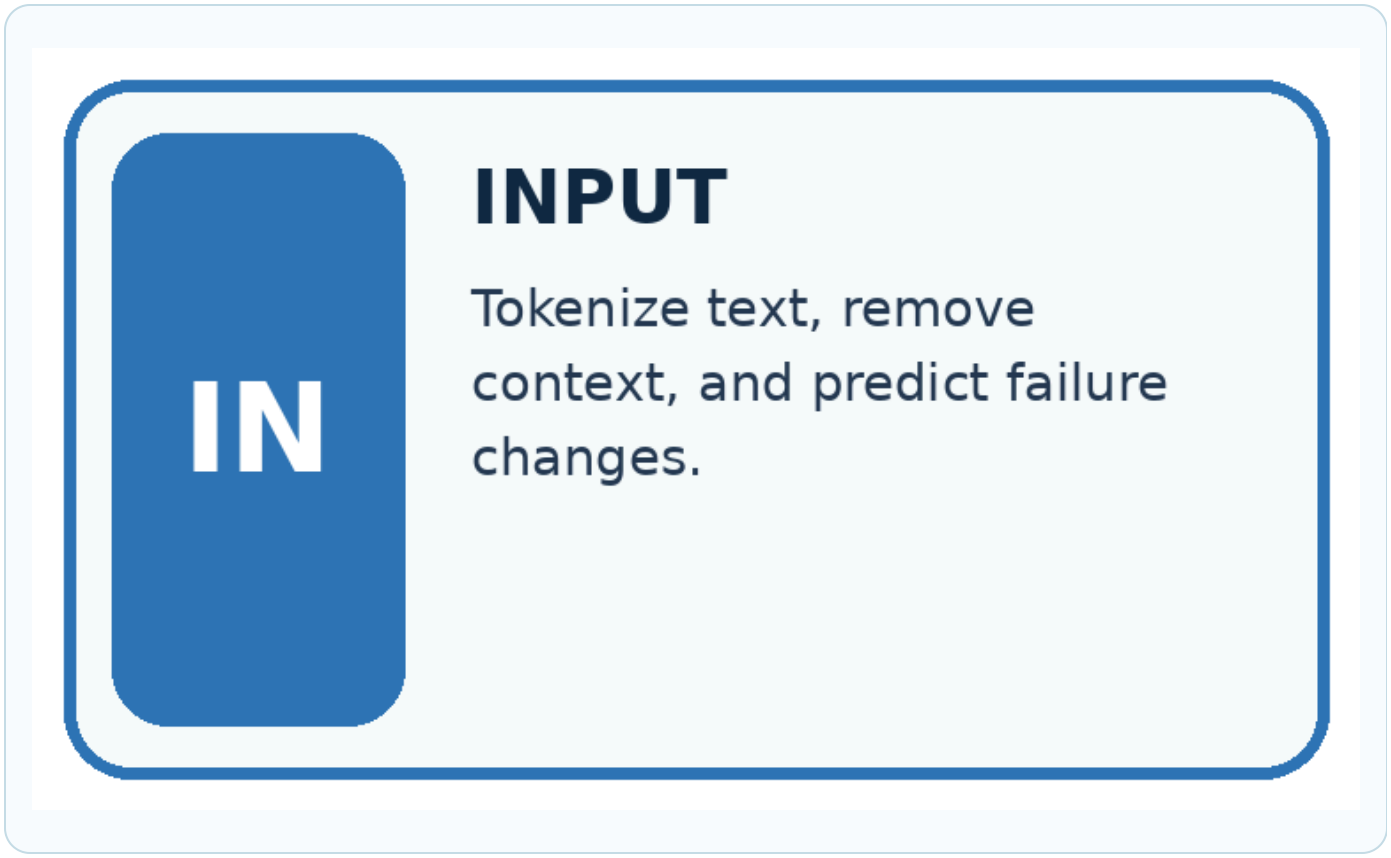
See how it works

> Follow the mechanism for Tokens and Context. Connect each labelled input, deciding step and result.

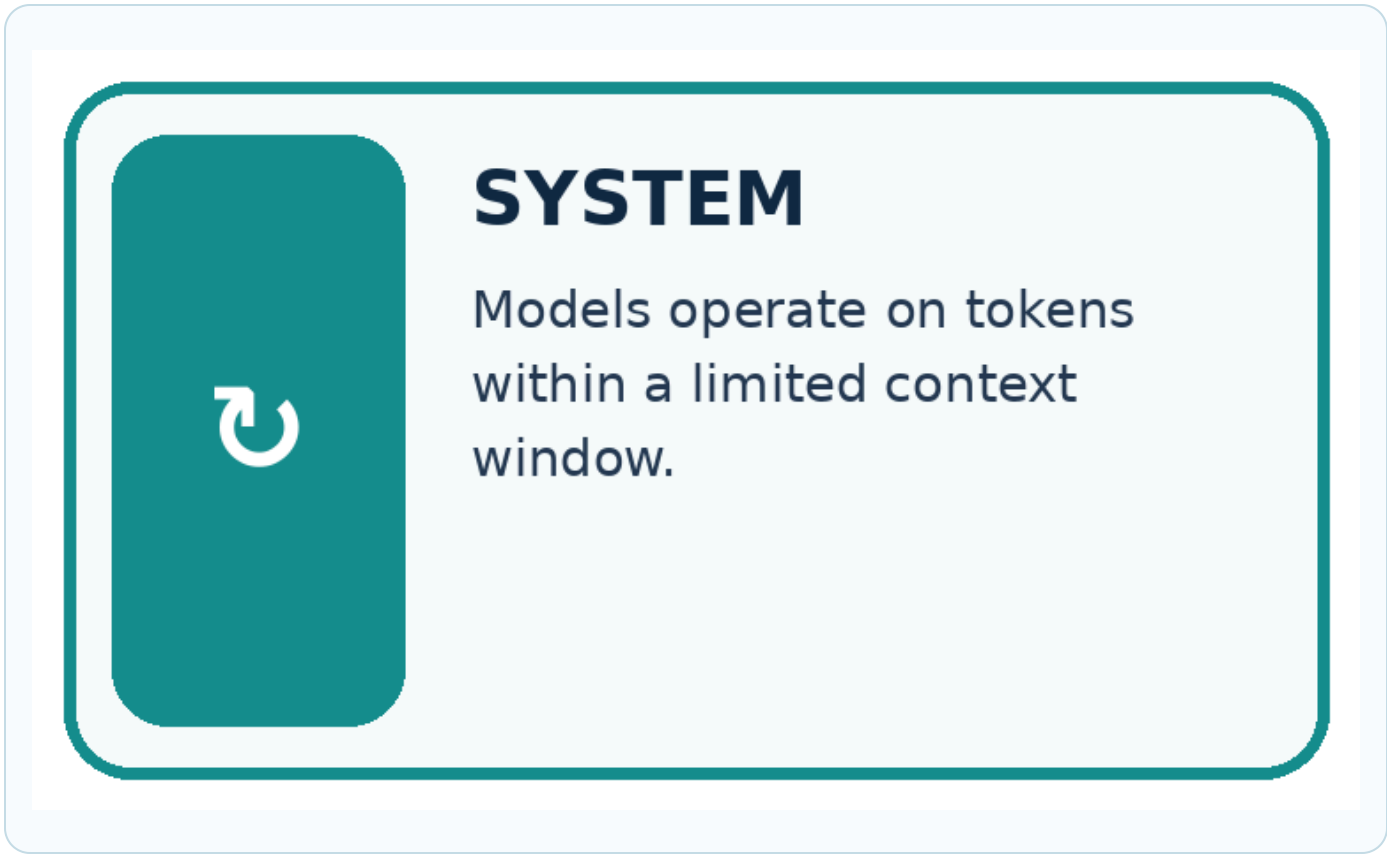
VISUAL MECHANISM

Input → Process → Output

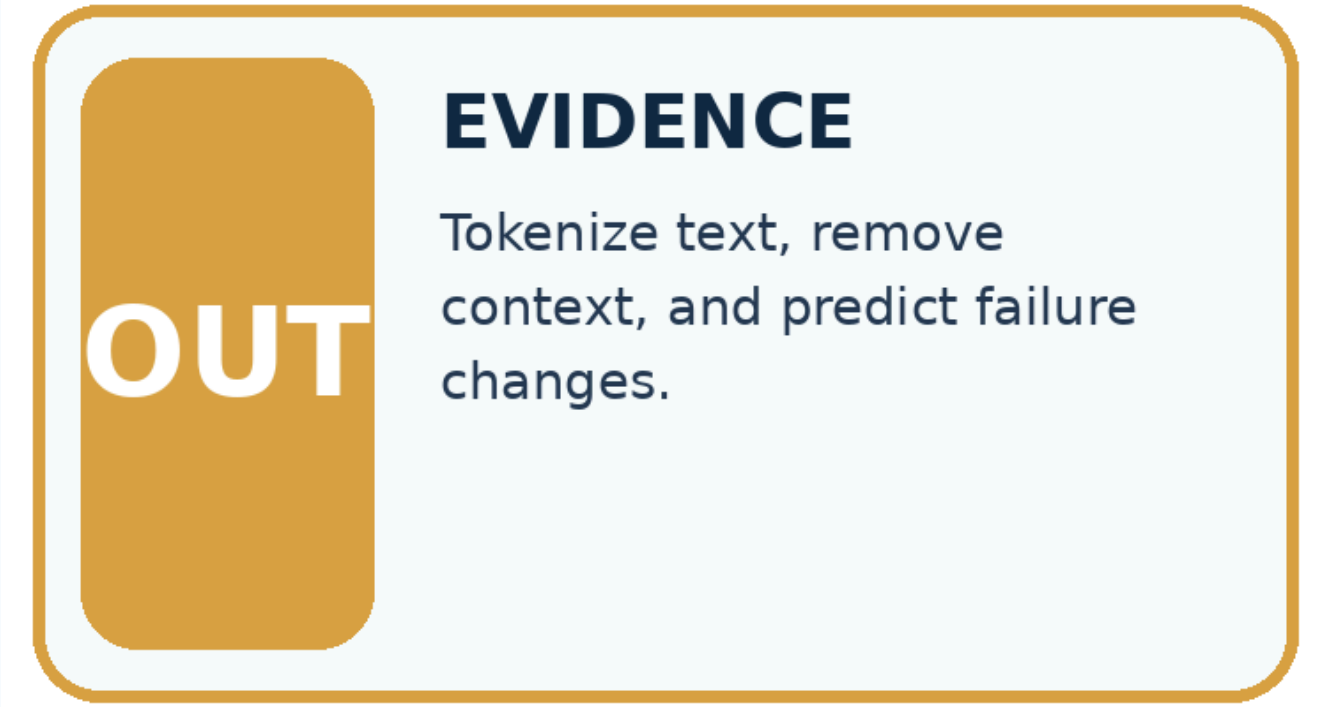
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Tokenize text, remove context, and predict failure changes.

↓


SYSTEM
Models operate on tokens within a limited context window.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Evidence	Cost	What it supports
A: current approval	30	Approved room B
B: current register	20	Room B capacity 12
C: superseded approval	25	Old room A approval

A+B costs 50 and fits exactly. Together they support both requested claims. A+C costs 55 and exceeds the **evidence** budget, while C+B fits at 45 but supplies an outdated room decision.

A cheap set is not necessarily an adequate set. Preserve document IDs, versions and dates alongside content.

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
100 total units	Subtract 20+10+20	50 evidence units
A 30 plus B 20	Exactly 50 evidence units	Supports both requested claims
Insufficient evidence	Ask or abstain	Do not invent the missing fact

Arithmetic fit and evidential adequacy are separate checks.

Apply the connected idea: Attention as weighted information selection

Value	Weight	Weighted vector
[2,0]	0.5	[1,0]
[0,4]	0.25	[0,1]
[4,2]	0.25	[1,0.5]

Adding the weighted vectors gives [2,1.5]. The same weights applied to different **value**s can give a different result; the same **value**s with different weights can also change it. For example weights 0,1,0 yield V2=[0,4].

The result is a representation to be used by later operations, not automatically a generated word.

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
Values with equal width	Multiply by .5, .25, .25	Contributions [1,0], [0,1], [1,.5]
Weighted contributions	Add corresponding coordinates	Output [2,1.5]
Mask excludes V3	Renormalise V1 and V2	Output [4/3,4/3]

The mask changes the eligible positions and therefore the probability mass assigned to the remaining values.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

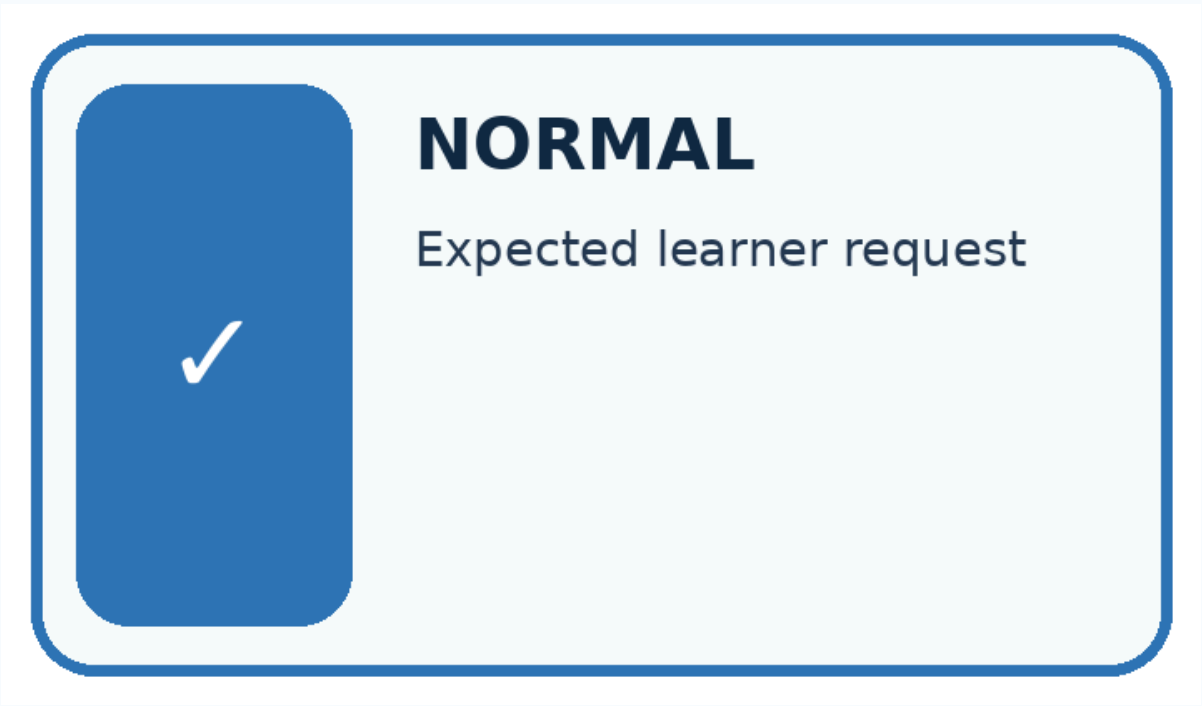
> Compare the supplied cases for Tokens and Context. Identify the detail that changes the conclusion.

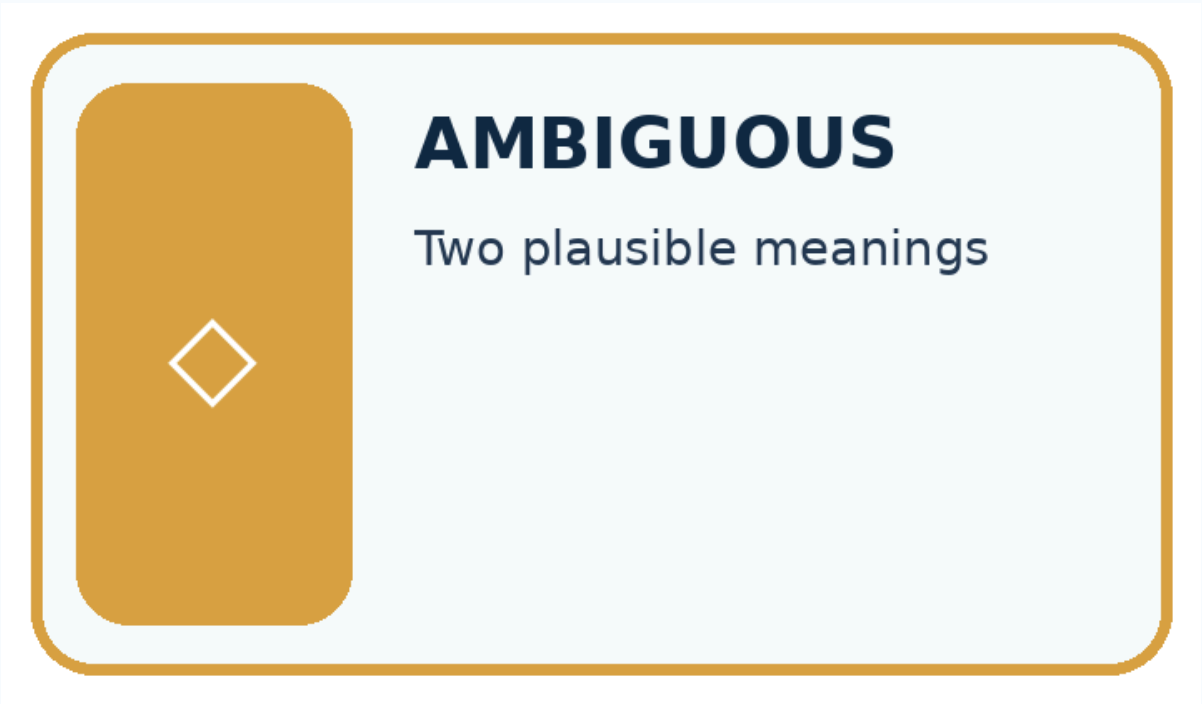
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

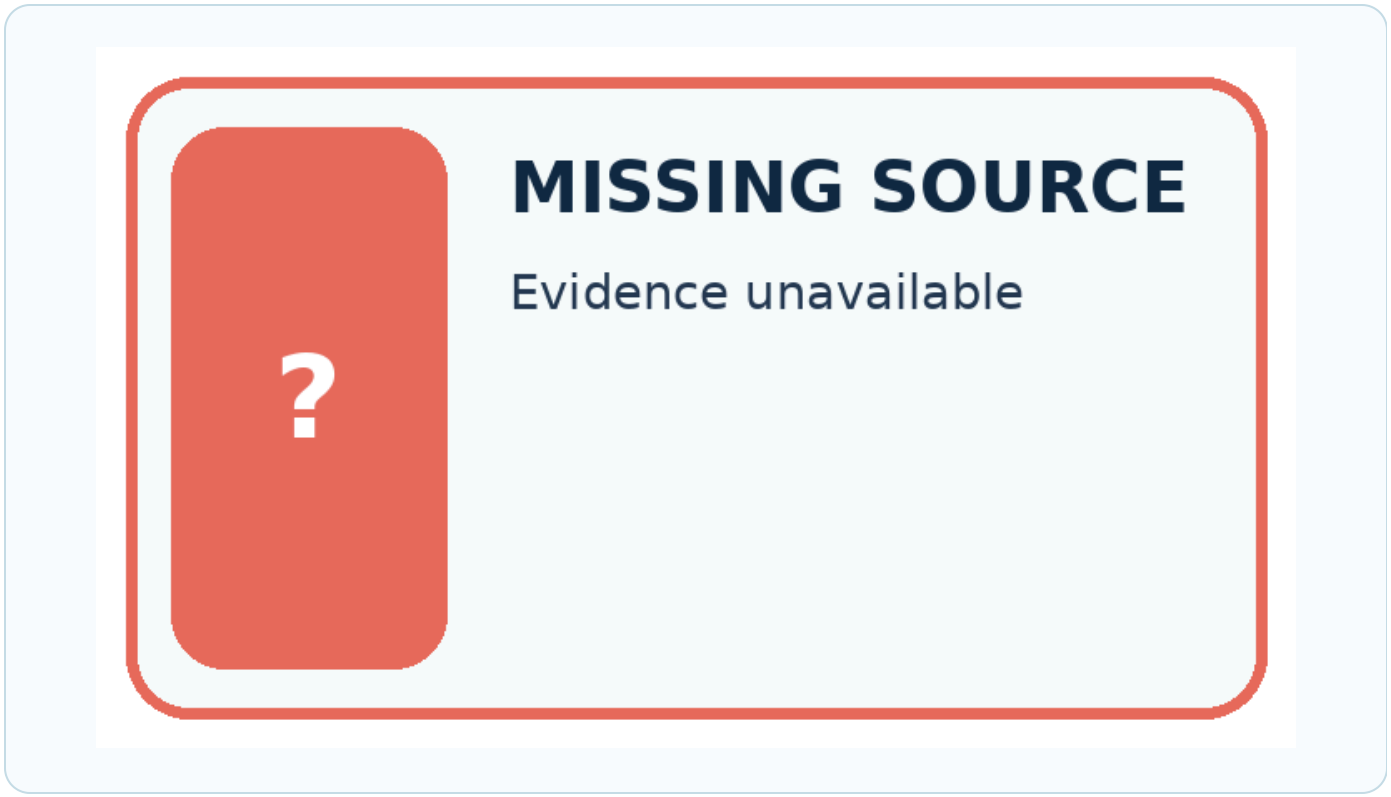
A system is not understood until it survives more than one comfortable example.

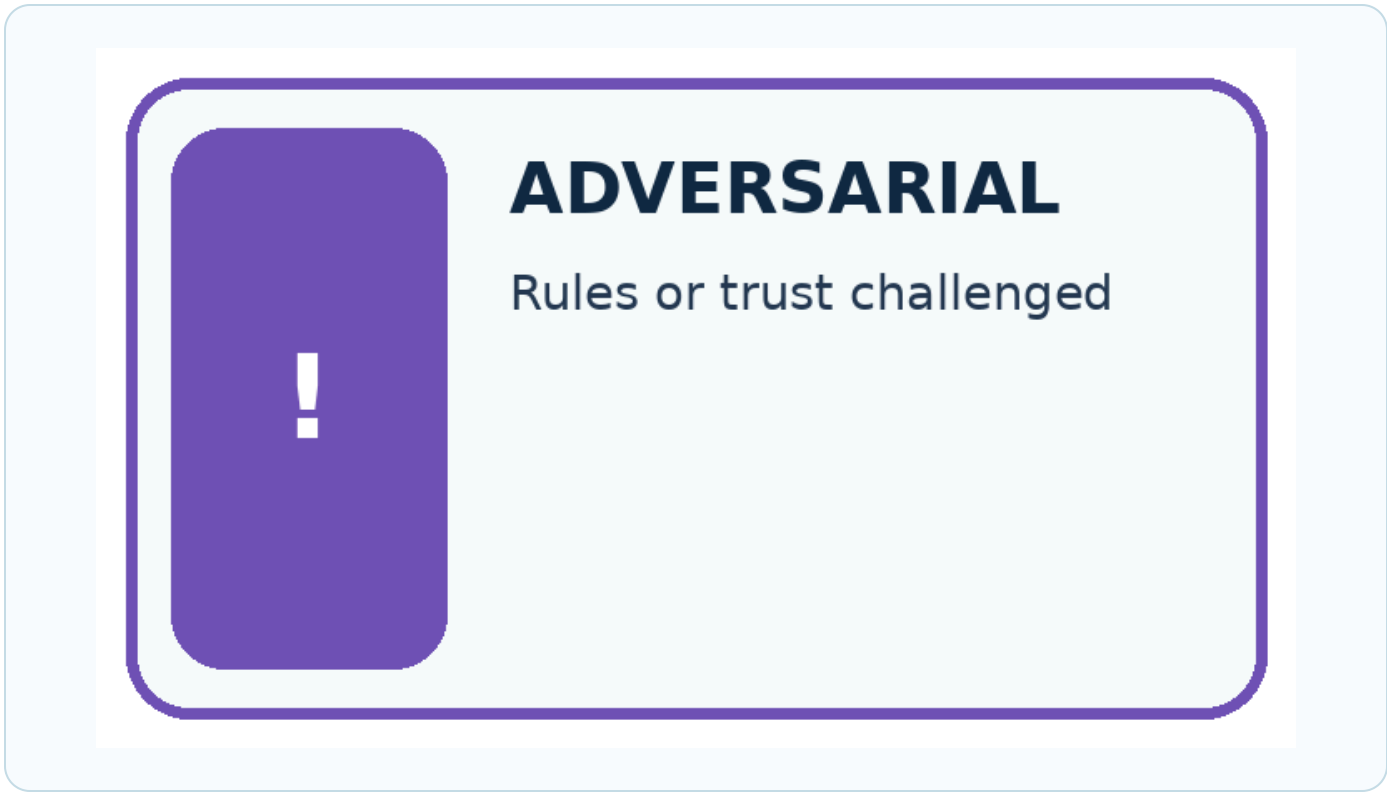




The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.


?
MISSING SOURCE
Evidence unavailable


!
ADVERSARIAL
Rules or trust challenged

Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Increase B's cost to 30 with all other costs fixed.

Predicted result

Prediction: A+B costs 48 and no longer fits the 45-**token evidence** capacity.

Why the result follows

Reason: Do not silently omit the exception; use an approved source-preserving summary, another retrieval step or an honest incomplete-**evidence** response.

STUDENT LESSON · 5 OF 18

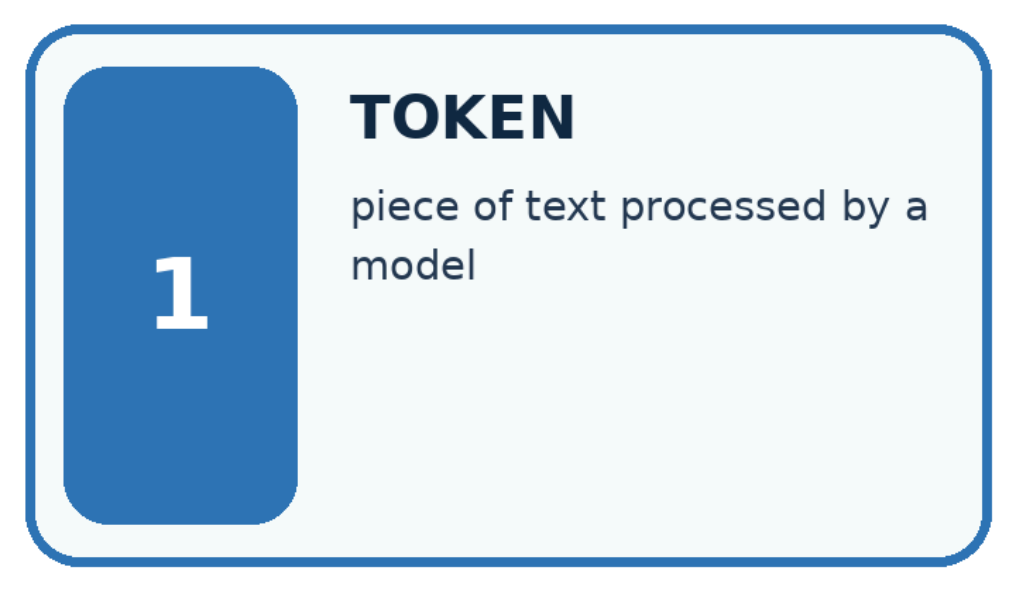
Words that unlock the idea

> Use the vocabulary of Tokens and Context to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



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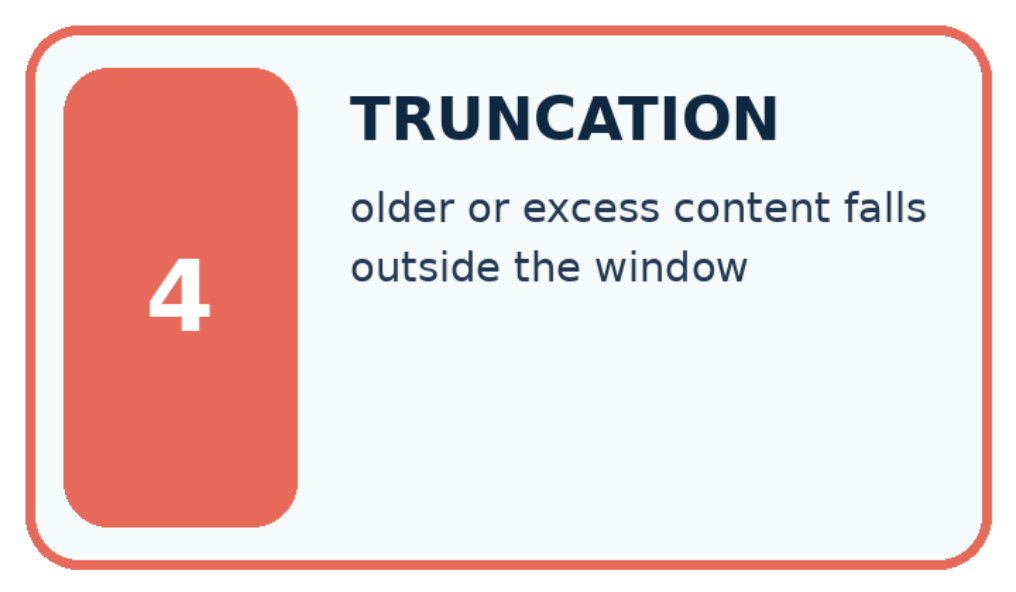
TOKENIZATION

method for splitting text into pieces

3

CONTEXT WINDOW

tokens available for the current response



Word	My own example	Not this / boundary
token	A unit of encoded text processed by a language model, which may be a word, part of a word or punctuation.	
tokenization	Converting text into the token units of a particular tokenizer.	
context window	The bounded amount of token information a model can process together in a particular use.	
truncation	Removing part of information, often because it exceeds a length or context limit.	

Words used in the closer explanation

Word	Meaning
Context budget	The permitted input and output allocation under a stated model or classroom contract
Evidence selection	Choosing source material relevant to the requested claim
Reserved output	Capacity deliberately kept available for the response

Terms for Attention as weighted information selection

Term	Meaning
Query	A vector used to compare available keys for the current calculation
Key	A vector used to determine compatibility with a query
Value	A vector combined using the resulting attention weights
Mask	A rule excluding specified positions from a calculation

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Tokens and Context, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 3 • WORKED EVIDENCE

1. A sentence becomes token pieces, not a stored human thought.
2. The model uses visible tokens to predict the next token.
3. Remove the definition from context: the same question becomes ambiguous.
4. More context helps only when it is relevant, accurate, and within the window.

1. A sentence becomes **token** pieces, not a stored human thought.

2. The model uses visible **token**s to predict the next **token**.

3. Remove the definition from context: the same question becomes ambiguous.

4. More context helps only when it is relevant, accurate, and within the window.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Worked explanation · Tokens and Context

A source may contain an exception after a general statement. Cutting off the last paragraph can reverse its practical meaning.

Selection should preserve the passages needed to interpret the requested fact, including scope, version and exceptions.

If complete relevant **evidence** cannot fit, retrieve a smaller authoritative passage or request clarification rather than silently remove a qualifying condition.

Including a document in context does not prove the generated answer uses it correctly. After generation, check each claim against its source.

A source's instruction-like content also does not acquire authority just because it fits in the **evidence** budget.

Apply the connected idea: Attention as weighted information selection

Worked explanation · Tokens and Context

Suppose V3 is a future position unavailable to the current autoregressive step. The **mask** excludes it. For this supplied-weight example, renormalise the permitted weights 0.5 and 0.25 by their total 0.75, giving 2/3 and 1/3.

The result becomes [4/3,4/3]. Merely setting V3's contribution to zero while keeping the old weights produces [1,1], whose retained weights sum to 0.75; that is not the specified normalised combination.

Worked explanation · Tokens and Context

If every position is excluded, the permitted total is zero. This contract reports no available **value** rather than dividing by zero or silently using a future position.

A real implementation needs its own documented handling for empty **mask**s. A causal **mask** restricts information paths; it does not make retrieved content factually trustworthy.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Tokens and Context

A fictional assistant has a small **context budget** for answering an event-capacity question.

The facts we will use

Supplied facts

Total budget 100 toy **token**s; instructions 20, question 10, **output** reserve 25. **Evidence** capacity is the remainder.

Supplied facts

Current capacity record A costs 18; its safety exception B costs 20; unrelated slogan C costs 12. A says 20 seats; B caps this event at 12.

AI ACADEMY / AI-3/3

Context budget allocations

	Toy tokens
Instructions	20
Question	10
Output reserve	25
Evidence available	45

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-3/3

Tokens and Context

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute budget

100-20-10-25=45 evidence tokens.



2. Select relevant pair

A+B costs 38 and preserves the capacity exception.



3. Reject incomplete shortcut

A+C costs 30 but leaves out the exception that changes the answer.



4. Answer with scope

For this event, the declared cap is 12, with both A and B retained.

CHANGE ONE THING

Increase B's cost to 30 with all other costs fixed.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute budget	100−20−10−25=45 evidence tokens.
2. Select relevant pair	A+B costs 38 and preserves the capacity exception.
3. Reject incomplete shortcut	A+C costs 30 but leaves out the exception that changes the answer.
4. Answer with scope	For this event, the declared cap is 12, with both A and B retained.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **context window** limits how much information is available in one model call. Selecting the shortest passage is not always sufficient: an exception can change the meaning of a general statement.

Keep the relevant source relationship visible. Toy **token** costs are supplied here; real **token** counts depend on the model's **token**izer.

What this example does not establish

Keep the stated limit

Fitting in context does not guarantee attention, correct interpretation or accurate generation.

Keep the stated limit

Read the labelled picture: Can A alone justify answering 20 seats after the exception is known?

Keep the stated limit

Change one thing: Increase B's cost to 30 with all other costs fixed. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Tokens and Context claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A token is always exactly one English word. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

If room approval is missing, do not infer it from capacity. If sources conflict without a documented current version, report the conflict or ask the responsible person.

Keep the requested claim, budget calculation, selected source IDs and unresolved facts in the record. This classroom accounting develops the reasoning without needing a live model.

Exit: A is unavailable and B gives only room capacity. Write a response that reports the supported fact without claiming the meeting is approved there.

Apply the connected idea: Attention as weighted information selection

Check the claim against the facts

A large weight means a large coefficient in this particular weighted sum. It is not a complete explanation of the whole network, which may contain other heads, transformations and residual paths.

It also does not establish that the high-weight source is true. Trace the actual **value**s and downstream use before making a broader claim.

Exit: a team says a 0.9 attention weight proves a document is correct and allows following its instructions. Explain the two unsupported inferences.

STUDENT LESSON · 8 OF 18

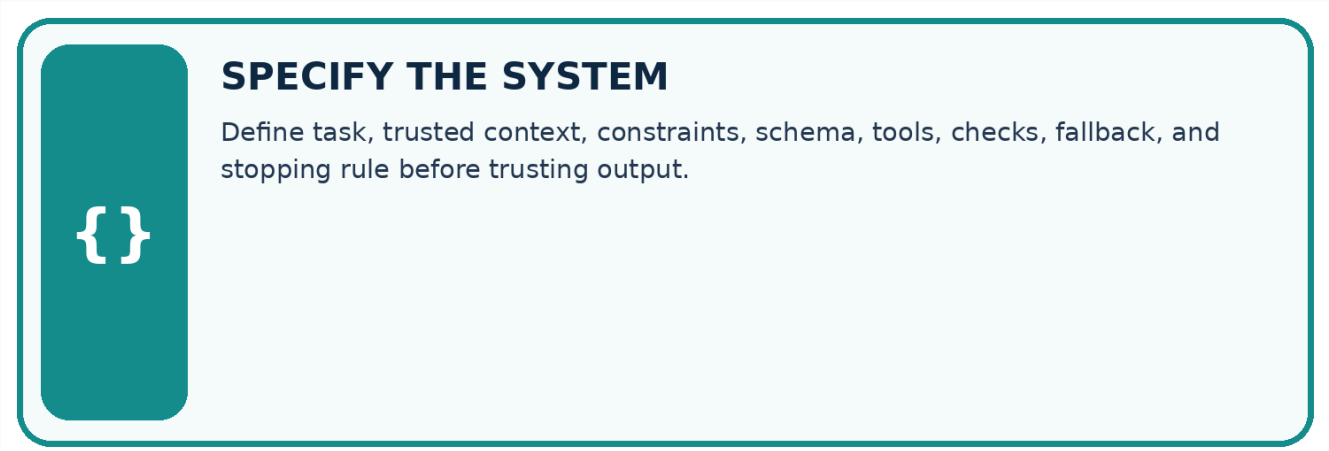
Practise together

> Practise Tokens and Context with support, then explain the deciding step in your own words.

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Supported total 80, fixed instructions 15, question 10 and **reserved output** 15 leave 40 units. Two complete relevant passages cost 18 and 22, **fitting** exactly.

A third irrelevant passage costs 8; adding it would exceed the stated total. A short summary can reduce size but must retain claim-supporting qualifiers and be checked against the original.

Guided: calculate the available evidence units and explain why exact fit does not prove answer correctness.

Apply the connected idea: Attention as weighted information selection

Supported **value**s A=[6,0], B=[0,3], weights 1/3 and 2/3. Their weighted combination is [2,2]. **Mask**ing B leaves A with normalised weight 1 and result [6,0].

This is a different information set, not a numerical repair to the original result.

Guided: compute both results and explain why a masked-out source must not influence the combination.

Practise with less help: Choose complete support

Use workbook W2 for the connected example “Context limits and **evidence selection**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Context limits and **evidence selection**. Proposed explanation: “**Token** count equals word count”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Attention as weighted information selection

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: Attention copies the highest-weight **token**

Explain the picture

> Can A alone justify answering 20 seats after the exception is known?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

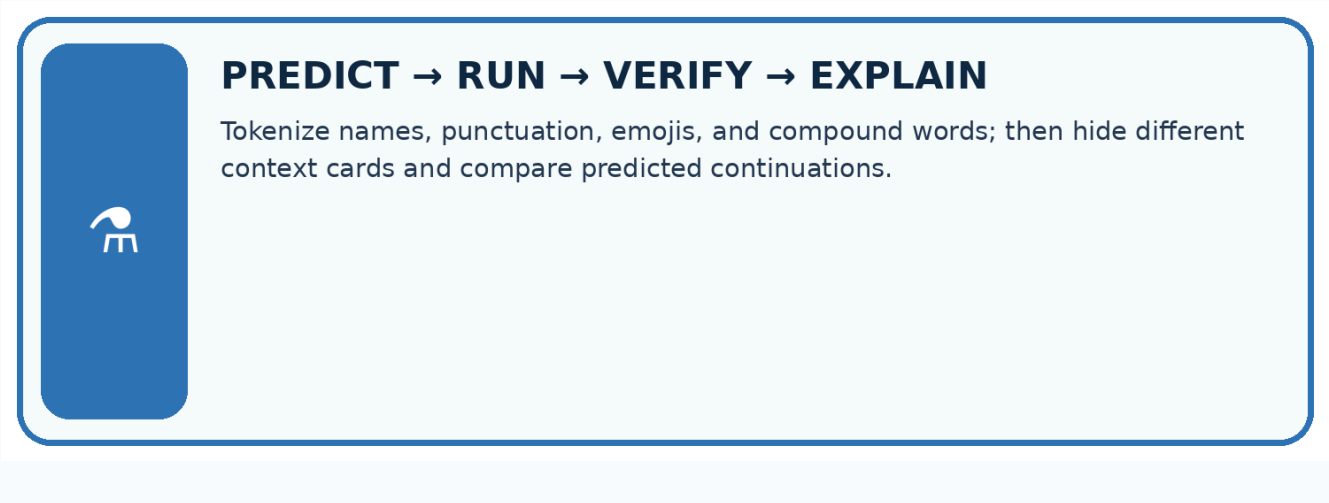
Run the investigation

> Investigate Tokens and Context. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Tokenize names, punctuation, emojis, and compound words; then hide different context cards and compare predicted continuations.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

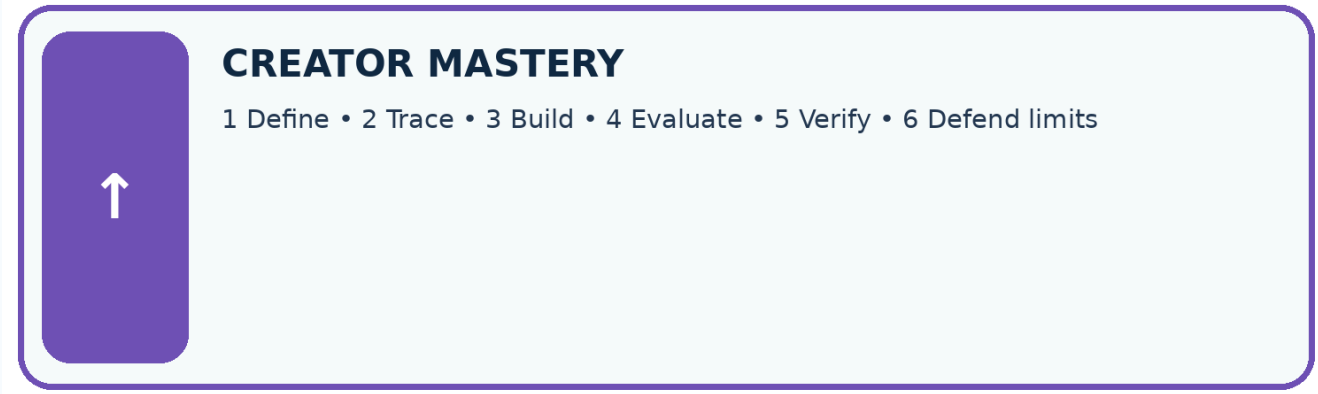
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Tokens and Context. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

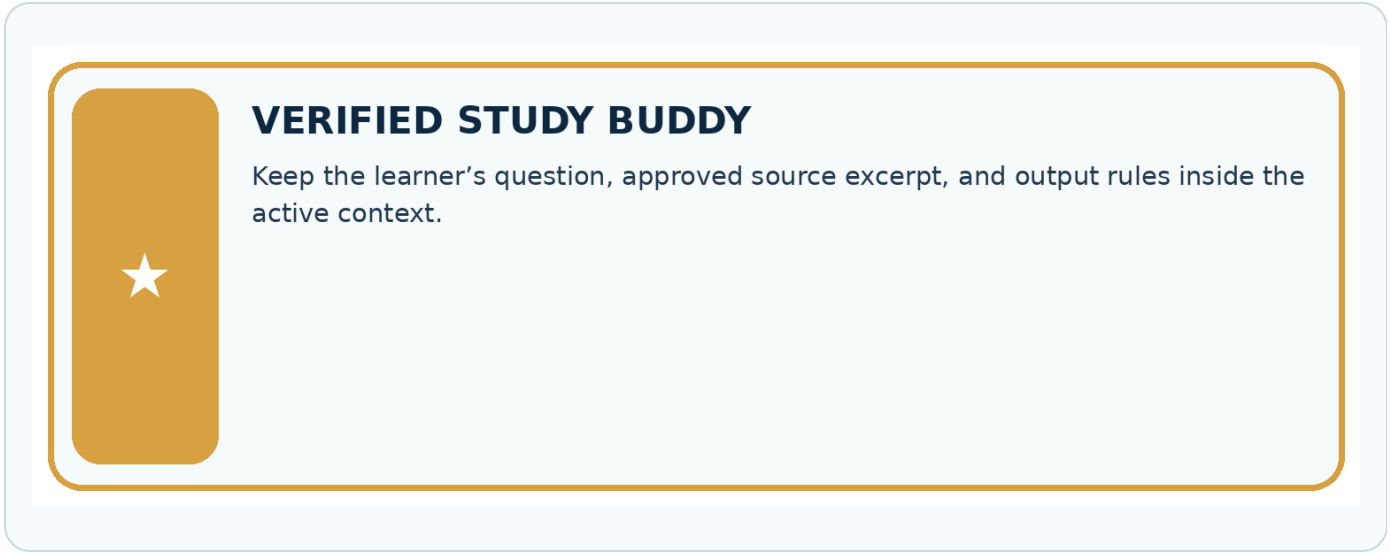
Connect your project

> Apply this week's idea to Verified Study Buddy. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK’S CHECKPOINT Keep the learner’s question, approved source excerpt, and **output** rules inside the active context.

Trace the system

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

Evidence for your project

Design Study Buddy retrieval to preserve current facts, exceptions and source identity within its **context budget**.

STUDENT LESSON · 12 OF 18

Show what you know

Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.

✓

INDEPENDENT CREATOR PROOF

Tokenize text, remove context, and predict failure changes.

Try a more demanding case

CHALLENGE **Token**ize text, remove context, and predict failure changes.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Tokens and Context case for supported practice. Repeated examples do not become fresh assessments.

Week 3 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A toy window holds eight **token**s; prompt uses six and quotation uses three.

The question

> Does it fit? What should be preserved?

Worked explanation

Nine exceeds eight. Shorten or select material while preserving essential instructions and **evidence**. Real systems also have **output**-budget rules.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Nine requested tokens exceed eight spaces

Evidence or stage	Value or result
Prompt	6
Quotation	3
Capacity	8

A toy window holds eight **token**s; prompt uses six and quotation uses three.

> Does it fit? What should be preserved?

Nine exceeds eight. Shorten or select material while preserving essential instructions and **evidence**. Real systems also have **output**-budget rules.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Budget new evidence independently

Total 120, instructions 25, question 15, **output** reserve 30. Source D is current approval, cost 28; E is current capacity, cost 22; F is old approval, cost 18.

Compute capacity, compare D+E and E+F, and state an honest fallback if D is missing.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 3 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> A system can use only the last two messages. Message 1 says use centimetres; messages 2 and 3 discuss plants. What instruction may be lost, and how can the next request repair this?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Keep the learner’s question, approved source excerpt, and **output** rules inside the active context.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A toy system allows 12 **token**s total for **input** and **reserved output**. Instructions use 3, a source uses 7 and the planned answer needs 4. Compute the excess and propose a revision that preserves **evidence**.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

Use this map to navigate Tokens and Context, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional assistant has a small **context budget** for answering an event-capacity question.

By the end, I can	How I will show it
Account for fixed text, evidence and reserved output under a toy budget	Explain the deciding step, show a trace or test, and state one limit.
Select complete evidence for a stated question	Explain the deciding step, show a trace or test, and state one limit.
Handle insufficient or conflicting evidence explicitly	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Tokens and Context without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Account for fixed text, **evidence** and **reserved output** under a toy budget

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional assistant has a small **context budget** for answering an event-capacity question.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? For this event, the declared cap is 12, with both A and B retained.

1. Answer with scope

2. Select relevant pair

3. Reject incomplete shortcut

4. Compute budget

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Answer with scope For this event, the declared cap is 12, with both A and B retained. A **context window** limits how much information is available in one model call.

Selecting the shortest passage is not always sufficient: an exception can change the meaning of a general statement. Keep the relevant source relationship visible.

Toy **token** costs are supplied here; real **token** counts depend on the model's **token**izer.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Increase B's cost to 30 with all other costs fixed. Which conclusion is supported by the supplied facts?

1. A+B costs 48 and no longer fits the 45-**token evidence** capacity.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: A+B costs 48 and no longer fits the 45-**token evidence** capacity.

Do not silently omit the exception; use an approved source-preserving summary, another retrieval step or an honest incomplete-**evidence** response.

Repair a missing building block

Review A Distribution of Possibilities (ai-3/2). Goal: Check whether a next-**token** table is a valid probability distribution. Short practice: Why is 0.625 not a broken probability sum?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

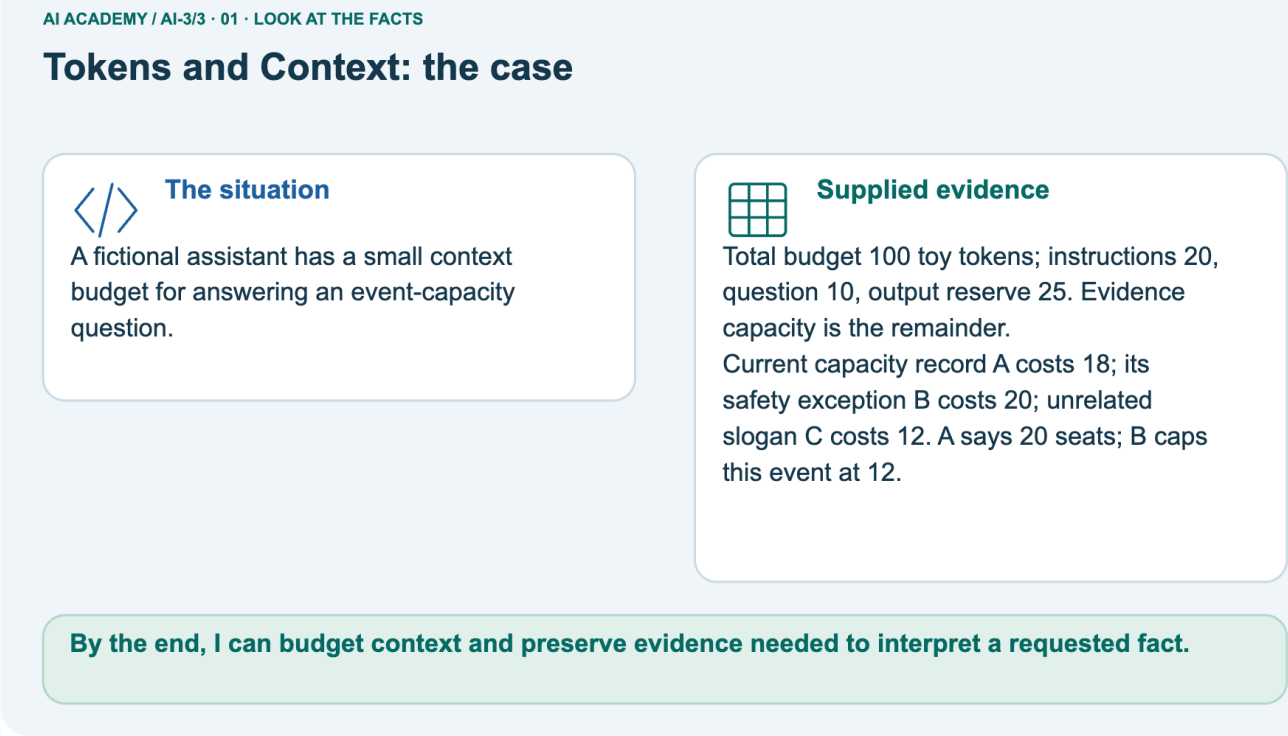
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Tokens and Context graphic using its labels, deciding evidence and relevant limitation.

01. Tokens and Context: the case



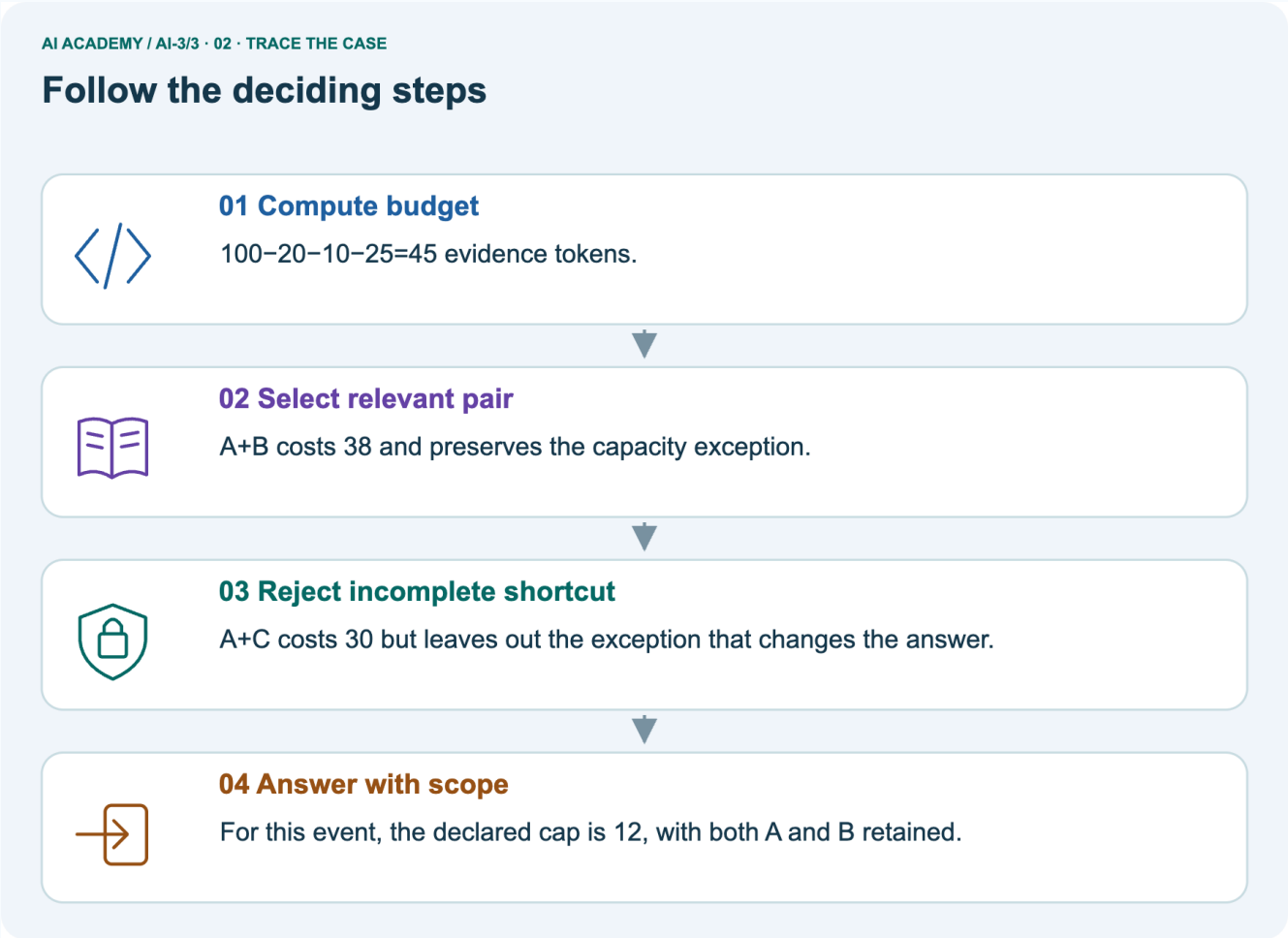
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional assistant has a small **context budget** for answering an event-capacity question. Supplied facts: Total budget 100 toy **token**s; instructions 20, question 10, **output** reserve 25. **Evidence** capacity is the remainder.

Current capacity record A costs 18; its safety exception B costs 20; unrelated slogan C costs 12. A says 20 seats; B caps this event at 12.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Compute budget: 100−20−10−25=45 **evidence token**s. → Select relevant pair: A+B costs 38 and preserves the capacity exception. → Reject incomplete shortcut: A+C costs 30 but leaves out the exception that changes the answer. → Answer with scope: For this event, the declared cap is 12, with both A and B retained.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Context budget allocations

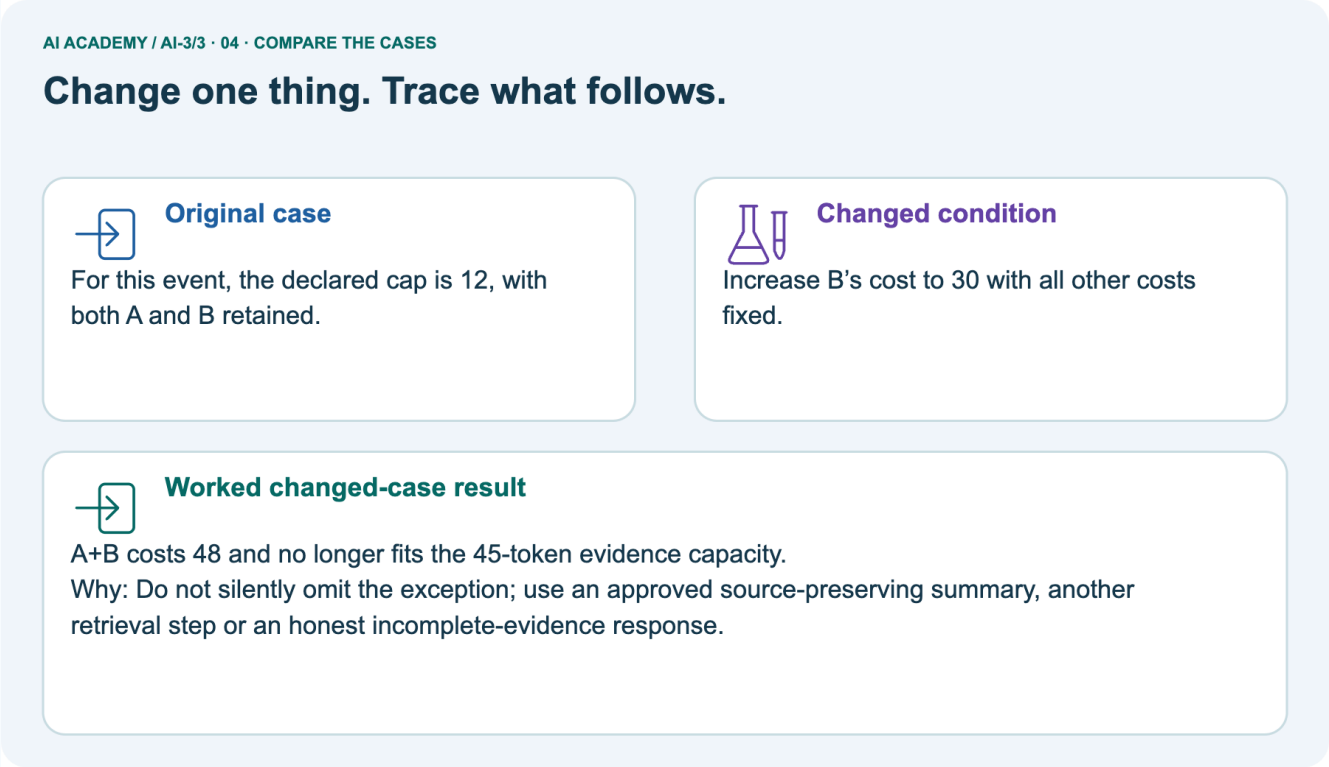


Follow the labelled picture with the matching explanation. Open the vector diagram

Context budget allocations. Instructions: Toy **token**s = 20. Question: Toy **token**s = 10. **Output** reserve: Toy **token**s = 25. **Evidence** available: Toy **token**s = 45.

Read the labels and **value**s. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

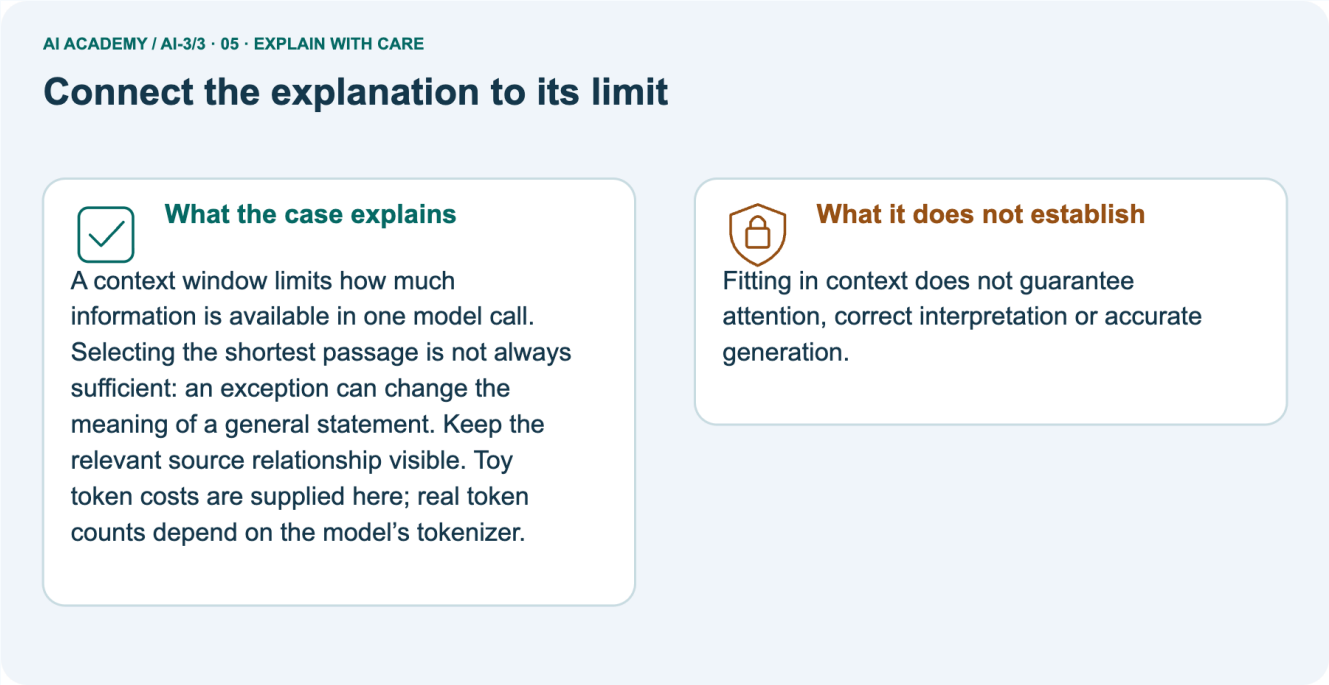
Original case: For this event, the declared cap is 12, with both A and B retained. Changed condition: Increase B's cost to 30 with all other costs fixed.

Expected result: A+B costs 48 and no longer fits the 45-**token evidence** capacity.

Reason: Do not silently omit the exception; use an approved source-preserving summary, another retrieval step or an honest incomplete-**evidence** response.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

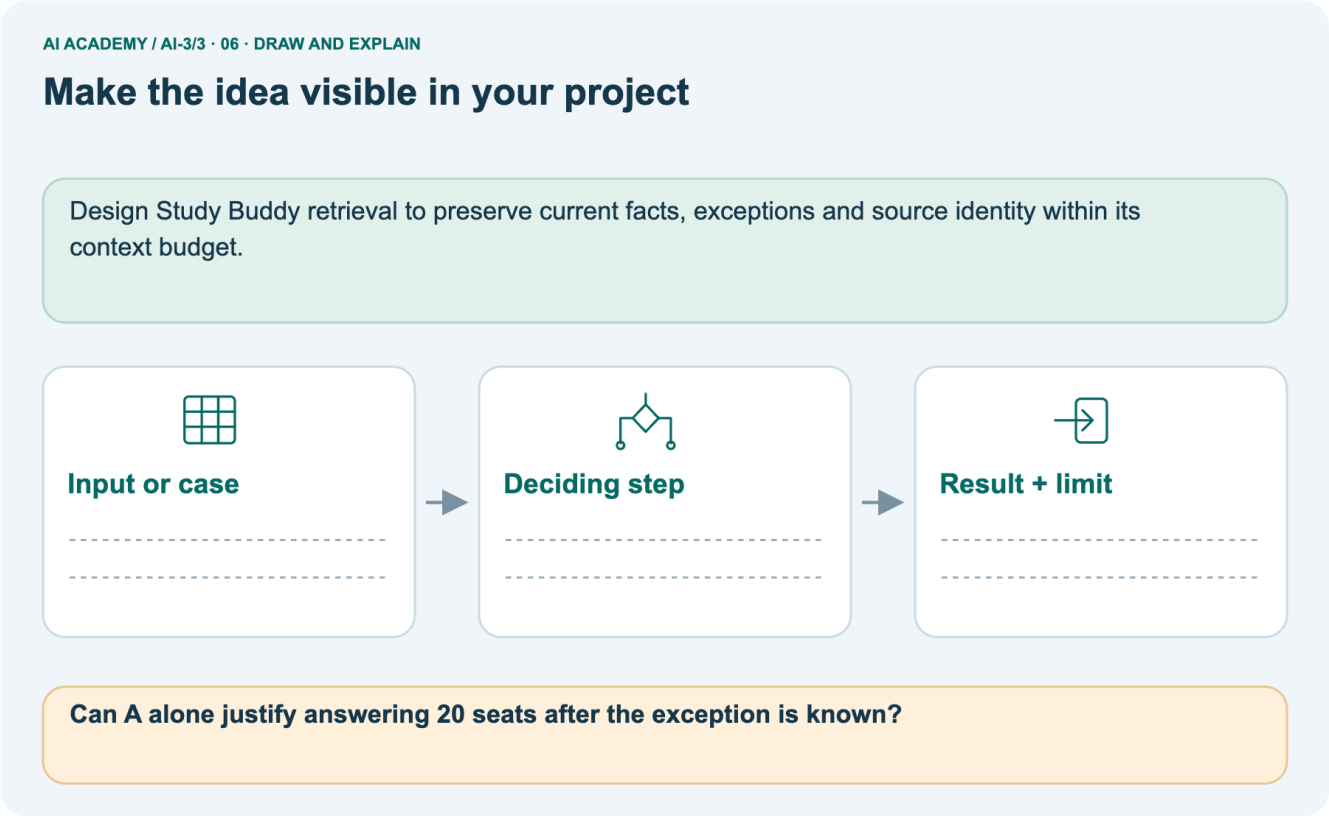
A **context window** limits how much information is available in one model call. Selecting the shortest passage is not always sufficient: an exception can change the meaning of a general statement.

Keep the relevant source relationship visible. Toy **token** costs are supplied here; real **token** counts depend on the model's **token**izer.

Limit: **Fitting** in context does not guarantee attention, correct interpretation or accurate generation.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Design Study Buddy retrieval to preserve current facts, exceptions and source identity within its **context budget**. Draw your **input**, deciding step, result and limit.

Supported question: Can A alone justify answering 20 seats after the exception is known?

Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.